

SWARUP KUMAR SUBUDHI

Reliability Engineering | Accelerated Testing & Qualification | Cross-Functional NPI

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PROFESSIONAL SUMMARY

PhD candidate in Mechanical Engineering (University of Maryland, expected Dec 2026) with 4+ years of manufacturing and NPI leadership at Tata Motors and 4+ years of federally sponsored hardware qualification research under NASA-GSFC. Experienced designing and executing accelerated test plans — thermal cycling, vibration/shock, and environmental stress screening — from concept through qualification, and driving PFMEA-based root-cause corrective action with cross-functional and supplier teams. Applies reliability methodology (Physics of Failure, Weibull analysis, Six Sigma) across electro-mechanical hardware domains.

EXPERIENCE

Graduate Research Assistant · University of Maryland, College Park

Jan 2022 – Present

Development of Additively Manufactured Hard X-Ray Polarimeter Assembly — Sponsor: NASA, GSFC

- Designed and executed qualification test plans for launch environments, including high-g vibration testing (launch-load profiles) and environmental stress screening; established pass/fail criteria and documented the full qualification matrix for program review
- Achieved $\pm 10\ \mu\text{m}$ print tolerances for high-density (50–100 μm) silver-ink circuits on space-grade substrates, enabling printed electronics qualified for launch environments
- Designed, fabricated, and integrated a rotary 2-DOF electro-mechanical fixture into an nScript 3Dn platform for high-density circuit printing on detector strips and read-out boards; led hardware-system integration, functional testing, and system code development
- Applied machine design fundamentals, GD&T, and tolerance stack-up analysis against ISO/ASME standards throughout hardware development

Thermal Cycling Reliability of Printed Electronics

- Ran accelerated thermal cycling (-40°C to 140°C) on multi-layered printed dielectric and conductive stacks and performed FEA (ANSYS/Abaqus) to predict failure loci from inelastic strain accumulation and creep in sintered silver and dielectric materials
- Validated FEA failure predictions against experimental SEM and cross-section microscopy — closing the loop between simulation and physical failure analysis

RF Printed Electronics Reliability

- Fabricated and environmentally qualified MIS capacitors and coplanar waveguides (DIW-printed dielectric/conductive pastes) for aircraft-environment use; characterized RF performance (S-parameters, reflection coefficients, efficiency) against qualification criteria

Senior Manager, Manufacturing – Transmissions · Tata Motors Ltd. CVBU, India

2019 – 2020

- Led a cross-functional Six Sigma team (NPI, QA, SCM, R&D) to reduce gearbox bearing warranty failures ~50% across the 24-MIS vehicle population
- Drove PFMEA-based quality controls and re-balanced takt time at bottleneck stations, improving gearbox assembly line efficiency ~12%
- Implemented IoT-based real-time loss monitoring and Ishikawa-driven corrective action plans; owned production schedule adherence for a 6-speed gearbox line (2 line managers, 45–50 operators)

Assistant Manager, NPI – Transmissions · Tata Motors Ltd. CVBU, India

2016 – 2019

- Owned 3 OEM products end-to-end from design to field deployment, including DFA/DFM reviews, parts qualification, PLM integration, and test matrix development
- Launched auxiliary transmission systems for TATA Kestrel 8×8 defense vehicles; achieved 100% on-time production adherence across 10 new gearbox launches for 6×6/8×8 defense trucks
- Executed value-engineering case studies on synchronizing parts and hydraulic tipping kits, realizing ₹4.53 Cr in annual savings for the LKP2518 tipper family

EDUCATION

Ph.D., Mechanical Engineering · University of Maryland, College Park, MD — Expected Dec 2026

GPA: 3.83; Key courses: Probabilistic PoF & Accelerated Testing, Failure Mechanisms, Thermal Analysis of Components

M.S., Mechanical Engineering · University of Maryland, College Park, MD — 2021

B.Tech., Mechanical Engineering · National Institute of Technology, Rourkela, India — 2016

RELEVANT TECHNICAL SKILLS

Reliability & Testing: Thermal Cycling, Vibration/Shock Loading, Environmental Stress Screening (ESS), Temp-Humidity Bias (THB), Failure Analysis, Physics of Failure (PoF), Acceleration Factors, Weibull Analysis, Functional/Qualification Testing

Statistical & Quality: Six Sigma, DoE, ANOVA, SPC (Cpk), PFMEA, DFA/DFM, NPI Process, PLM, BOM

Component Characterization: SEM, TMA/CTE, DMA, Microscopy, Surface Profiling, GD&T, MSA/Gauge R&R

CAD / CAE / CFD: SolidWorks, Creo, ANSYS Mechanical/Icepak, Abaqus, COMSOL Multiphysics, Fluent

Computation & Instrumentation: Python (CV2, image/point tracking), MATLAB (optimization, image processing), Arduino, sensor integration

RELEVANT PUBLICATIONS

- Development of Additively Manufactured Hard X-Ray Polarimeter Assembly and Reliability Analysis for Launch. In review, 2026.
- Janamanchi, P., Subudhi, S.K., Dasgupta, A., and Quinn, E. Thermal Cyclic Fatigue Reliability of Printed Hybrid Electronics (PHE). EuroSimE 2025.
- Zhao, B., Sivasankar, V.S., Subudhi, S.K., Dasgupta, A., and Das, S. Printed carbon nanotube-based humidity sensors deployable on surfaces of widely varying curvatures. ACS Applied Nano Materials, 6(2), 1459–1474, 2023.